

ExoHabitAI

Deployment Guide

AI-Powered Exoplanet Habitability Prediction

Field	Value
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Date	February 2026
Backend	Render (Free Tier)
Frontend	Vercel (Free Tier)
Backend URL	https://exohabitai-7335.onrender.com
Frontend URL	https://exohabitai-frontend-nu.vercel.app
Total Cost	\$0 / month
Document	Deployment

1. Project Overview

ExoHabitAI is an AI-powered web application that predicts exoplanet habitability using a Random Forest model trained on NASA data. It has three components:

- Flask REST API — Python backend that handles prediction requests
- React + Vite — frontend user interface
- Random Forest model (.pkl) — loaded automatically by the backend on startup

Component	Technology	Platform	Cost
Backend	Flask + Gunicorn	Render (Free)	\$0
Frontend	React + Vite	Vercel (Free)	\$0
ML Model	scikit-learn RF	Inside Backend	\$0

2. Pre-Deployment Checklist

Before deploying, confirm all of the following:

- `models/random_forest.pkl` exists and is committed to GitHub
- `requirements.txt` includes `flask-cors`
- Procfile contains: `web: gunicorn --chdir backend app:app --bind 0.0.0.0:$PORT`
- `Predict.tsx` uses `VITE_API_URL` environment variable (not hardcoded `localhost`)
- `frontend/package.json` has: `"engines": { "node": "24.x" }`
- All changes are pushed to GitHub

3. Backend Deployment — Render

Render hosts the Flask API and the ML model. The model is loaded automatically when the backend starts.

Step 1 — Create Account

1. Go to <https://render.com>
2. Sign up with your GitHub account

Step 2 — Create Web Service

3. Click New → Web Service
4. Connect your GitHub repository
5. Fill in the following settings:

Field	Value
Environment	Python
Build Command	<code>pip install -r requirements.txt</code>
Start Command	<code>gunicorn --chdir backend app:app</code>
Instance Type	Free (\$0/month)

6. Click Create Web Service and wait for the build to complete

Step 3 — Verify

Once the status shows Live, open the backend URL in your browser:

<https://exohabitai-7335.onrender.com>

You should see: `{"message": "ExoHabitAI backend is running."}`

4. Frontend Deployment — Vercel

Vercel hosts the React frontend as a static site. It is faster than Render for frontends and never spins down.

Step 1 — Create Account

7. Go to <https://vercel.com>
8. Sign up with your GitHub account

Step 2 — Import Project

9. Click Add New → Project
10. Import your GitHub repository
11. Fill in the following settings:

Field	Value
Root Directory	frontend
Framework Preset	Vite
Build Command	npm run build
Output Directory	dist

Step 3 — Add Environment Variable

Before deploying, scroll down to Environment Variables and add:

Field	Value
Key	VITE_API_URL
Value	https://exohabitai-7335.onrender.com

Important: The .env file is ignored by Vercel. This variable must be added manually in the dashboard. After adding it, redeploy for it to take effect.

Step 4 — Verify

12. Open your Vercel URL: <https://exohabitai-frontend-nu.vercel.app>
13. Go to the Predict page
14. Enter sample values: Radius 1.0, Mass 1.0, Period 365, Temp 5778
15. Click Predict Habitability — should return a result with confidence score

5. ML Model

The model is not deployed separately. It lives inside the backend on Render. When the backend starts, app.py loads the .pkl file automatically:

```
BASE_DIR = os.path.dirname(os.path.abspath(__file__))
MODEL_PATH = os.path.join(BASE_DIR, '..', 'models', 'random_forest.pkl')
model = joblib.load(MODEL_PATH)
```

To verify the model file is committed to GitHub:

```
git ls-files models/
```

It should show: models/random_forest.pkl. If missing, force-add it:

```
git add -f models/random_forest.pkl
git commit -m "add model file" && git push
```

6. Running Locally

Open two terminals:

Terminal 1 — Backend

```
cd ExoHabitAI
python backend/app.py
```

Expected: Running on http://0.0.0.0:5000

Terminal 2 — Frontend

```
cd ExoHabitAI/frontend
npm run dev
```

Expected: Local: http://localhost:5173

Local .env (frontend/.env)

```
VITE_API_URL=http://localhost:5000
```

7. Keeping Backend Awake

Render's free tier spins down after 15 minutes of inactivity. The first request after sleeping takes 30-60 seconds. To keep it awake:

16. Go to <https://cron-job.org> and sign up free
17. Click Create Cronjob
18. Set URL: <https://exohabitai-7335.onrender.com>
19. Set interval: Every 10 minutes and Save

This pings the backend every 10 minutes so it never sleeps. Always open the backend URL once before a demo to confirm it is awake.

8. Common Errors & Fixes

Error	Fix
ModuleNotFoundError: flask_cors	Add flask-cors to requirements.txt and push to GitHub
Model file not found	Run: git add -f models/random_forest.pkl then push
HTTP 502 on Render	Backend is sleeping — open backend URL in browser to wake it up
Prediction shows Connection Error	VITE_API_URL missing in Vercel dashboard — add it and redeploy
Vercel build fails (Node version)	Add engines: { "node": "24.x" } to frontend/package.json

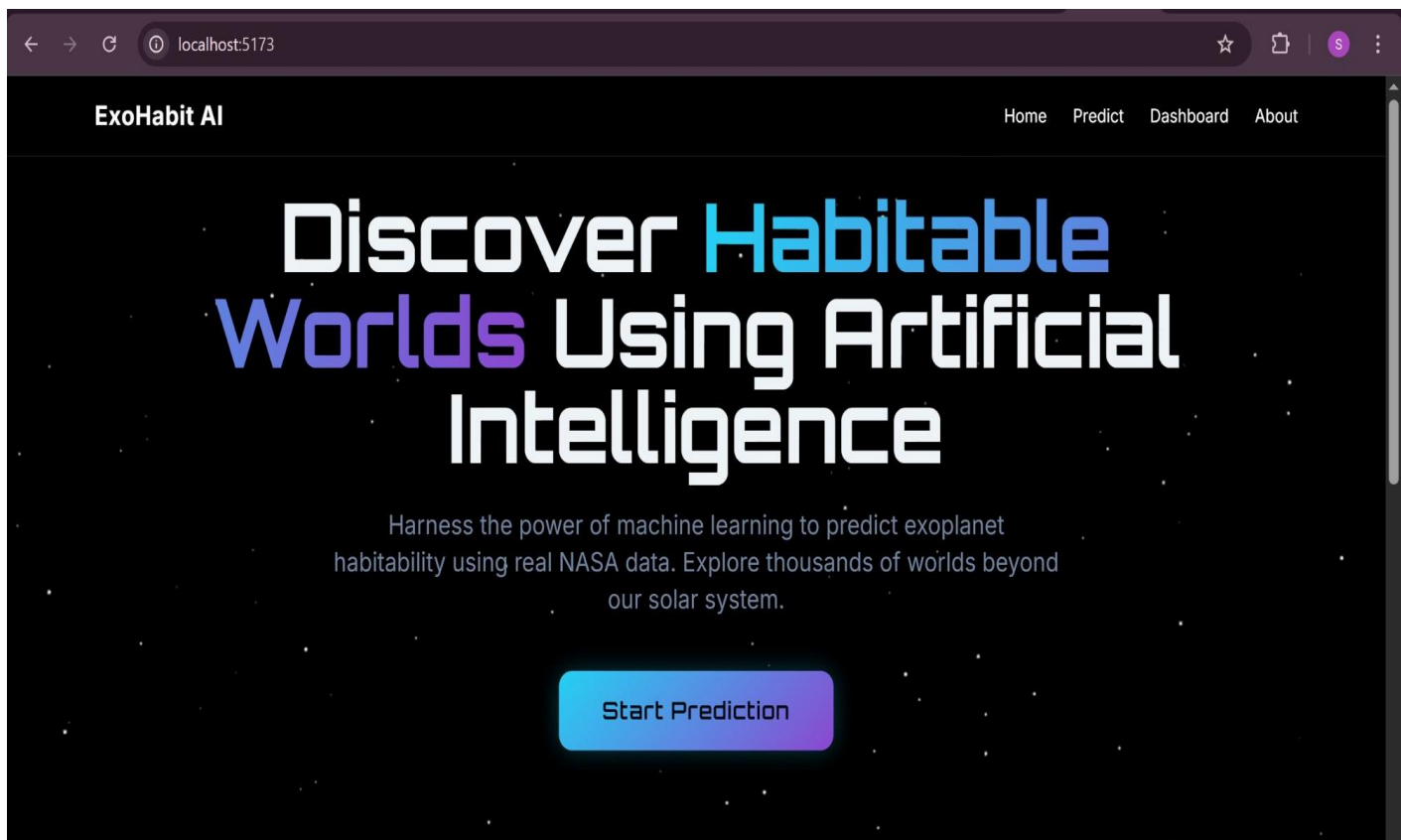
9. Verification Checklist

- Backend URL loads and returns running message
- Frontend URL loads and shows the home page
- Predict page returns a result with confidence score
- Dashboard page shows charts
- No errors in browser console (F12)

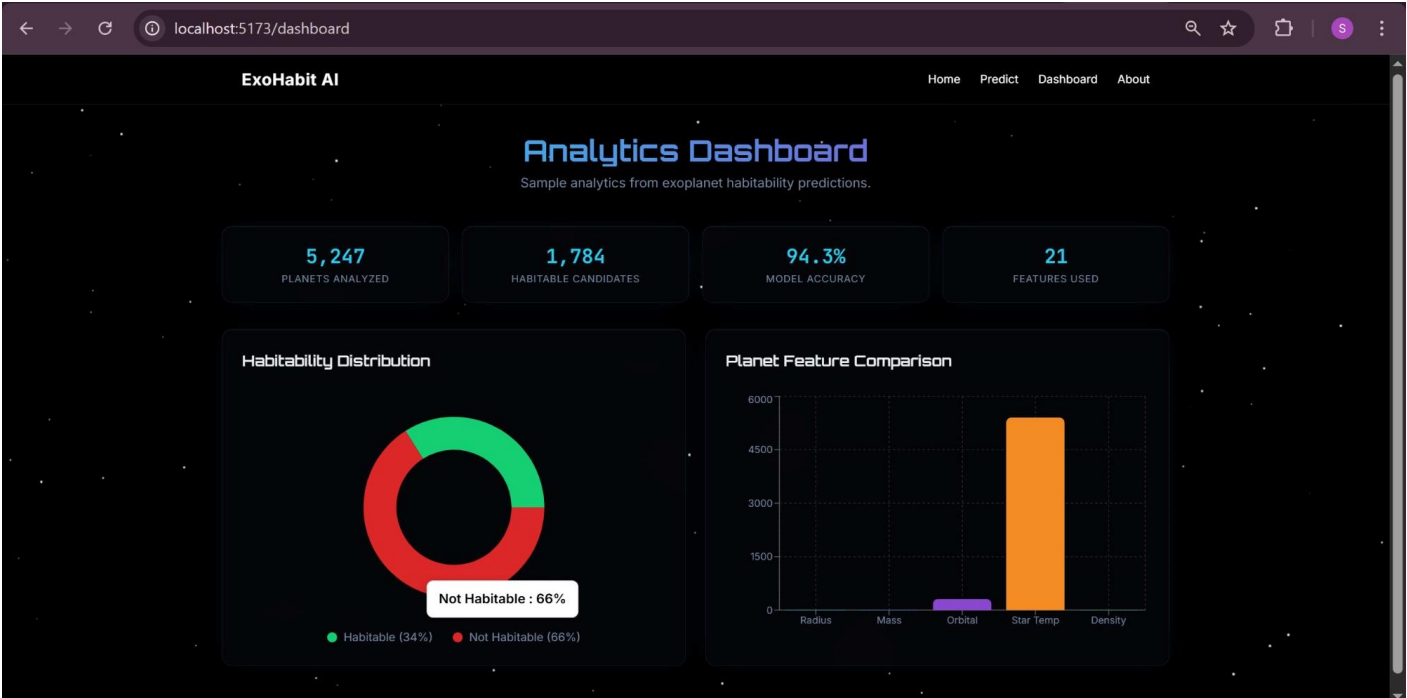
10. Screenshots of Working Web App

The following screenshots confirm the application is fully functional and all pages are working correctly.

Home Page — Landing screen with Start Prediction button



Dashboard — Analytics with habitability distribution chart



Predict Page — Input form with prediction result (56% Habitable)

The "Habitability Prediction" page allows users to input planetary parameters to predict habitability. The input form includes four fields: Planet Radius (Earth radii) set to 1, Planet Mass (Earth masses) set to 3, Orbital Period (days) set to 3, and Star Temperature (Kelvin) set to 7. A "Predict Habitability" button is located below these fields. The prediction result is displayed in a green-bordered box, showing a "+ HABITABLE" status, the text "This planet may support life!", and a "Confidence Score" of 56% represented by a green circular progress indicator.

Parameter	Value
Planet Radius (Earth radii)	1
Planet Mass (Earth masses)	3
Orbital Period (days)	3
Star Temperature (Kelvin)	7

Predict Habitability

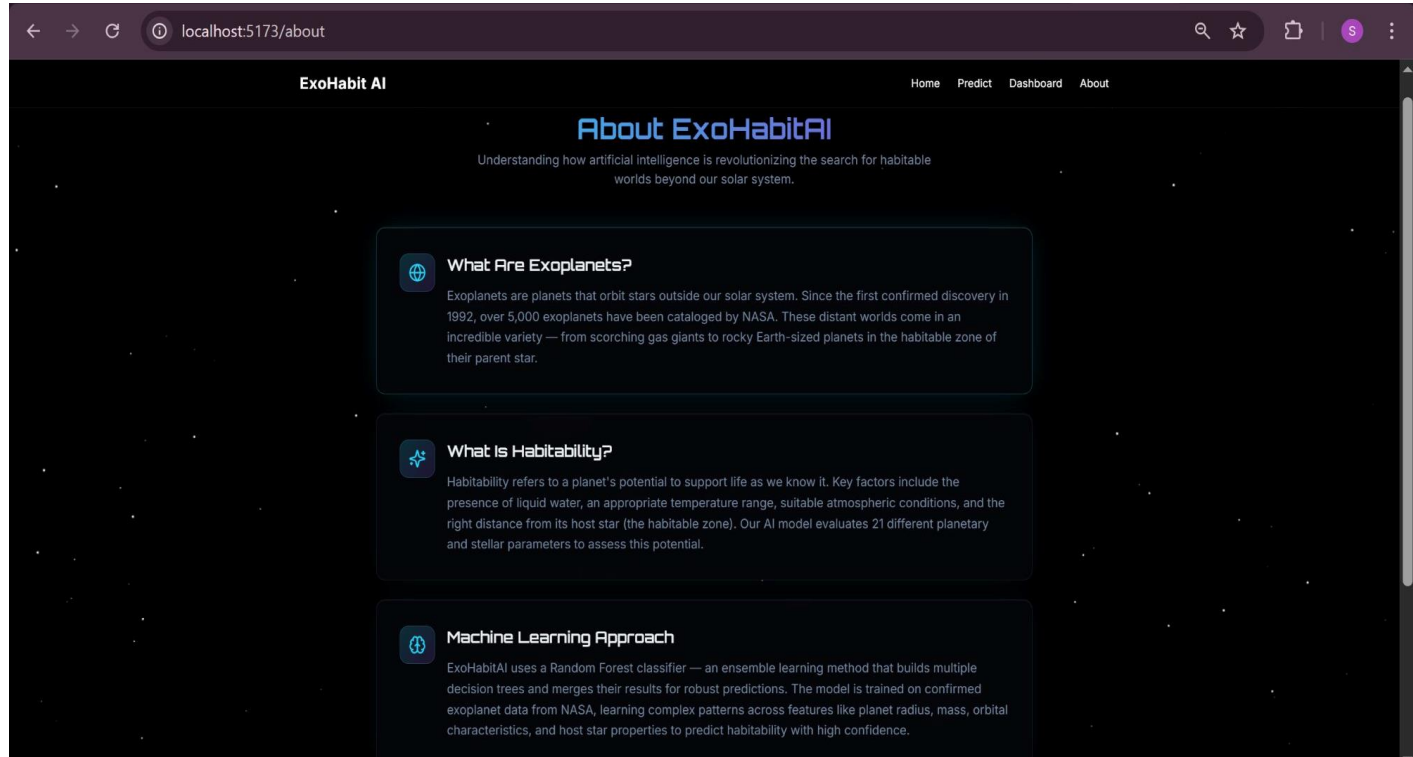
+ HABITABLE

This planet may support life!

56%

Confidence Score

About Page — Project information and ML approach



11. Deployment Links

All live URLs and resources for this project:

Live Frontend (Vercel): <https://exohabitai-frontend-nu.vercel.app>

Backend API (Render): <https://exohabitai-7335.onrender.com>

GitHub Deploy Repository: <https://github.com/rathodsakshi290/ExoHabitAI>

Project Github Repository: https://github.com/GKSJ19/ExoHabitAI/tree/Sakshi_Rathod